Spectral functions via novel cumulant expansions

(Dated: November 6, 2026)

I. INTRODUCTION

Electron-boson interactions are ubiquitous in condensed matter physics. For example, in core spectroscopy, the injection of a localized charge carrier couples to the plasmonic modes of the system, forming a fermionic polaron. This manifests in the observed spectrum as a sharp quasiparticle peak accompanied by satellite structures separated by the plasma frequency. In order to facilitate the interpretation of experiments, one may compute the spectral function, which is the direct observable in spectroscopy experiments.

Many-body perturbation theory (MBPT) is commonly used to obtain the spectral function, which is proportional to the imaginary part of the retarded single-particle Green's function G. MBPT obtains the Green's function through perturbative expansion in the bosonic propagator; in electronic structure, it is common to choose bosonic propagator to be the screened Coulomb interaction W. Doing so has the function of improving upon the mean-field answer at reasonable computational cost. The lowest-order term in this perturbative expansion is G_0W_0, which has been empirically successful at describing quasiparticles in systems with weak to intermediate coupling. However, G_0W_0 famously fails to describe satellite features. One might hope to remedy this by incorporating self-consistency into the perturbative expansion. However, self-consistent GW does not necessarily yield a better description of the satellites than G_0W_0; this has been attributed to the neglect of vertex corrections in GW.

Coupled-cluster theory, a nonperturbative framework which has been established as the gold standard of ground-state molecular quantum chemistry when up to triple excitations are considered (CCSDT), offers a route to these vertex corrections. Excited-state properties, including spectral functions, may be obtained from its equation-of-motion extension (EOM-CC). However, the steep computational scaling of EOM-CCSDT limits its application to small supercells. Although one may slightly reduce the scaling by truncating the excitation hierarchy at doubles, it is unclear to what extent this incorporates the desired vertex corrections.

The cumulant expansion presents a viable alternative, as it incorporates a class of vertex corrections by resumming the perturbation series. The cumulant expansion was identified as the exact solution for a single fermion coupled to a bath of bosons. As a result, it qualitatively reproduces the core spectrum, among other features. The cumulant expansion is typically performed on top of a G_0W_0 calculation, an approach we refer to as G_0W_0+C. Consequently, G_0W_0+C inherits the starting-point dependence of G_0W_0. Furthermore, studies of finite supercells of the electron gas have shown that G_0W_0+C deviates sizeably from numerically exact results. Therefore, we see a need to explore cumulant approaches apart from G_0W_0+C.

In Sec. II, after recounting G_0W_0+C, we introduce BWs2+C, a cumulant method based on a renormalized perturbation expansion in the bare Coulomb interaction v. We also study a self-consistent cumulant expansion. In Sec. **??**, we use the aforementioned methods to compute spectral functions. Finally, in Sec. **??**, we discuss our findings and and propose future directions.

II. THEORY

In what follows, $i, j, \ldots$ and $a, b, \ldots$ denote momentum states below and above the Fermi level, respectively, while $p, q, \ldots$ denote general momentum states. We work in the spin-restricted formalism. For a momentum k, all cumulant methods determine the time-domain single-particle Green's function $G_k(t)$ as

$$G_k(t) = G_k^0(t)\, e^{C_k(t)}, \tag{1}$$

with $G_k^0(t)$ the non-interacting Green's function and $C_k(t)$ the cumulant.

A. BWs2+C

In the BWs2 partition,

$$G_k^{(0)}(\omega) = \Big(\underbrace{\omega - \varepsilon_k^{\mathrm{HF}}}_{\left(G_k^{\mathrm{HF}}(\omega)\right)^{-1}} - \underbrace{(\alpha_1 \mathbf{W}^O \oplus \alpha_2 \mathbf{W}^V)_{kk}}_{W_{kk}} \Big)^{-1}, \tag{2}$$

where $\varepsilon_k^{\mathrm{HF}}$ is the HF single-particle energy of momentum state k. Elements of the $\boldsymbol{W}$ regularizer are $W_{ii}^O = \frac{1}{2}\Sigma_{ii}^{(2),\mathrm{2p1h}}[\tilde{\boldsymbol{\varepsilon}}](\omega = \tilde{\varepsilon}_i)$ and $W_{aa}^V = \frac{1}{2}\Sigma_{aa}^{(2),\mathrm{2h1p}}[\tilde{\boldsymbol{\varepsilon}}](\omega = \tilde{\varepsilon}_a)$. $\Sigma_{pp}^{(2),\mathrm{ch}}[\tilde{\varepsilon}](\omega)$ is the diagonal element of a given channel (ch=2h1p or ch=2p1h) of the second-order Born self-energy $\Sigma^{(2)}$ *evaluated at* the BWs2 single-particle energies $\tilde{\boldsymbol{\varepsilon}}$ (whose pth element is $\tilde{\varepsilon}_p$).

The interpretation is that in order to determine $G_k^{(0)}$ for BWs2+C, a self-consistent optimization must first be performed, that requires repeated evaluation of the second-order self-energy, which scales as $O(N^5)$.

The second Born expansion of the Dyson equation takes the form

$$G_k^{-1}(\omega) \approx \left(G_k^{\mathrm{HF}}(\omega)\right)^{-1} - \Sigma_{kk}^{(2)}[\boldsymbol{\varepsilon}^{\mathrm{HF}}](\omega). \tag{3}$$

The BWs2 partition, chooses a different partition, where the Dyson equation takes the form

$$G_k^{-1}(\omega) \approx \left(G_k^{(0)}(\omega)\right)^{-1} - \Sigma_{kk}^{\mathrm{eff}}(\omega), \tag{4}$$

where $\Sigma^{\text{eff}}_{kk}(\omega) \equiv \Sigma^{(2)}_{kk}[\tilde{\boldsymbol{\varepsilon}}](\omega) - W_{kk}$. We are motivated to see how the MBPT truncation of 4 fares for computing excited-states in the solid-state because the renormalization of single-particle energies in BWs2 is hypothesized to have the function of incorporating incorporates dynamical screening. Furthermore, it has been shown by Tew that BWs2 facilitates the accurate computation of quasiparticle energies for strongly correlated molecular systems. Note that the approximation in 4 remains exact up to second order in v. Equating the cumulant and Dyson expansions to second order in the bare Coulomb interaction gives

$$C_k(t) = i \int \frac{d\omega}{2\pi} e^{-i(\omega-\tilde{\varepsilon}_k)t} \frac{\Sigma^{\text{eff}}_{kk}(\omega)}{(\omega-\tilde{\varepsilon}_k)^2} \tag{5}$$

$$= i \int \frac{d\omega'}{2\pi} e^{-i\omega' t} \frac{\Sigma^{\text{eff}}_{kk}(\omega'+\tilde{\varepsilon}_k)}{(\omega')^2}. \tag{6}$$

For momentum $k < k_F$, we have

$$\Sigma^{\text{eff}}_{k}(\omega) = \Sigma^{(2)}_{kk}[\tilde{\boldsymbol{\varepsilon}}]\,(\omega) - \frac{\alpha_1}{2}\Sigma^{(2),\text{2p1h}}_{kk}[\tilde{\boldsymbol{\varepsilon}}]\,(\tilde{\varepsilon}_k) \tag{7}$$

$$\implies \Sigma^{\text{eff}}_{k}(\omega+\tilde{\varepsilon}_k) = \Sigma^{(2)}_{kk}[\tilde{\boldsymbol{\varepsilon}}]\,(\omega+\tilde{\varepsilon}_k) - \frac{\alpha_1}{2}\Sigma^{(2),\text{2p1h}}_{kk}[\tilde{\boldsymbol{\varepsilon}}]\,(\tilde{\varepsilon}_k) \tag{8}$$

$$\implies C_k(t) = \sum_{iab} \frac{\mathcal{K}^{ab}_{ki}}{\left(\tilde{\epsilon}^{ab}_{ki}\right)^2} \left(e^{-i\tilde{\epsilon}^{ab}_{ki}t} + i\,(1-\alpha_1/2)\,\tilde{\epsilon}^{ab}_{ki}t - 1\right) + \sum_{ija} \frac{\mathcal{K}^{ka}_{ij}}{\left(\tilde{\epsilon}^{ij}_{ka}\right)^2} \left(e^{-i\tilde{\epsilon}^{ij}_{ka}t} + i\tilde{\epsilon}^{ij}_{ka}t - 1\right), \tag{9}$$

where $\mathcal{K}^{rs}_{pq} = (pr|qs)\,[2(pr|qs) - (ps|qr)]$, $\tilde{\epsilon}^{ab}_{ki} = \tilde{\varepsilon}_a + \tilde{\varepsilon}_b - \tilde{\varepsilon}_k - \tilde{\varepsilon}_i$, and $\tilde{\epsilon}^{ij}_{ka} = \tilde{\varepsilon}_i + \tilde{\varepsilon}_j - \tilde{\varepsilon}_k - \tilde{\varepsilon}_a$. By analogy for momentum $k > k_F$, we have

$$C_k(t) = \sum_{iab} \frac{\mathcal{K}^{ab}_{ki}}{\left(\tilde{\epsilon}^{ab}_{ki}\right)^2} \left(e^{-i\tilde{\epsilon}^{ab}_{ki}t} + i\tilde{\epsilon}^{ab}_{ki}t - 1\right) + \sum_{ija} \frac{\mathcal{K}^{ka}_{ij}}{\left(\tilde{\epsilon}^{ij}_{ka}\right)^2} \left(e^{-i\tilde{\epsilon}^{ij}_{ka}t} + i\,(1-\alpha_2/2)\,\tilde{\epsilon}^{ij}_{ka}t - 1\right). \tag{10}$$

Note that this reduces to the second-order cumulant when $\alpha_1 = \alpha_2 = 0$.

B. GW+C

We compute spectral functions of the homogeneous electron gas at a Wigner–Seitz radius of $r_s = 4$. The electron gas is a bellwether system for excited-state methods, whose ultimate goal is to predict the spectral functions of real materials. This density is of practical interest because it coincides with the valence electron density of the simple alkali metals. It is also of theoretical interest because, at $r_s = 4$, the electron gas lies outside the perturbative regime, where there is a notable lack of existing methods that can accurately compute excited states.

The theories investigated below make use of the screened Coulomb interaction, in which the screening is obtained from the direct random-phase approximation (RPA). This amounts to solving the Casida equation

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ -\mathbf{B} & -\mathbf{A} \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \boldsymbol{\Omega}, \tag{11}$$

where

$$A_{ia,jb} = \delta_{ij}\delta_{ab}\left(\epsilon^{\text{LDA}}_a - \epsilon^{\text{LDA}}_i\right) + v(\mathbf{q}), \tag{12}$$

$$B_{ia,jb} = v(\mathbf{q}). \tag{13}$$

Here, i and j label bands, $a = i + \mathbf{q}$ and $b = j + \mathbf{q}$ label unoccupied orbitals, and $v(\mathbf{q}) = 4\pi/(Vq^2)$ is the Coulomb interaction in a supercell of volume V.

Because momentum is conserved, the Casida matrix is block diagonal in the transferred momentum $\mathbf{q}$. Diagonalizing each block yields a set of bosonic modes ν (which include the plasmon) with frequencies $\Omega_\nu(\mathbf{q})$. The corresponding hole–boson coupling is

$$g_\nu(\mathbf{k},\mathbf{q}) = \sqrt{2}\sum_{ia}(\mathbf{k}\,,\mathbf{k}-\mathbf{q}\,|\,i\,a)\,(\mathbf{X}+\mathbf{Y})_{ia,\nu} = \sqrt{2}\,v(\mathbf{q})\sum_{i}(\mathbf{X}+\mathbf{Y})_{i,\,i+\mathbf{q}} \tag{14}$$

where the second equality follows because, for plane waves, every allowed two-electron integral equals $v(\mathbf{q})$; the coupling is therefore independent of $\mathbf{k}$. A naive diagonalization of the full Casida matrix scales as the sixth power of system size. By exploiting momentum conservation, we instead diagonalize a set of much smaller, independent blocks, one per momentum sector $\mathbf{q}$; the cubic symmetry of the electron gas further reduces the number of symmetry-inequivalent sectors that must be treated.

Note that in the electron gas, the LDA exchange–correlation potential is a constant, so the LDA eigenvalues are simply the free-electron kinetic energies shifted by V^{xc}. The LDA starting point therefore contains no correlation in its dispersion.

C. One-shot GW+C

In the cumulant formalism, the Green's function is written as

$$G_k(t) = G^0_k(t)\,e^{C_k(t)}, \tag{15}$$

where $G^0_k(t)$ is a non-interacting reference Green's function built from ϵ^{LDA}_k, and $C_k(t) = \tilde{C}_k(t) - i\Lambda_k t$ is the cumulant. The static part of the cumulant is $\Lambda_k = \Sigma^{\text{x}}_k - V^{\text{xc}}_k$, where Σ^{x}_k is the exchange self-energy. The reference Green's function and the dynamic part $\tilde{C}_k(t)$ depend on whether one constructs the time-ordered or the retarded cumulant, as described below.

1. Time-ordered cumulant

In the time-ordered cumulant (TC), the dynamic part is

$$\tilde{C}_k^{\mathrm{T}}(t) = \int d\omega \, \frac{\beta_k^{\pm}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right), \tag{16}$$

where

$$\beta_k^{\pm}(\omega) = \frac{1}{\pi} \left|\mathrm{Im}\, \Sigma_k^{\mathrm{R}}\left(\omega + \epsilon_k^{\mathrm{LDA}}\right)\right| \theta\left(\pm\left(\omega + \epsilon_k^{\mathrm{LDA}} - \epsilon_{\mathrm{F}}\right)\right). \tag{17}$$

The non-interacting reference is

$$G_k^{\mathrm{T},0}(t) = \mp i\, \theta(\pm t)\, \theta\left(\pm\left(\epsilon_k^{\mathrm{LDA}} - \epsilon_{\mathrm{F}}\right)\right) e^{-i\epsilon_k^{\mathrm{LDA}} t}. \tag{18}$$

Throughout, the upper (lower) sign refers to particles with $k > k_{\mathrm{F}}$ (holes with $k < k_{\mathrm{F}}$), $\epsilon_{\mathrm{F}} \equiv \epsilon_{k_{\mathrm{F}}}^{\mathrm{LDA}}$ is the Fermi energy, and θ is the Heaviside step function. Thus, for a hole, only the portion of the self-energy below the Fermi level contributes to the cumulant.

2. Retarded cumulant

In the retarded cumulant (RC), the dynamic part is

$$\tilde{C}_k^{\mathrm{R}}(t) = \int d\omega \, \frac{\beta_k^{\mathrm{R}}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right), \tag{19}$$

where

$$\beta_k^{\mathrm{R}}(\omega) = \frac{1}{\pi} \left|\mathrm{Im}\, \Sigma_k^{\mathrm{R}}\left(\omega + \epsilon_k^{\mathrm{LDA}}\right)\right|, \tag{20}$$

so that both the particle and hole portions of the self-energy contribute. The non-interacting reference always takes the form

$$G_k^{\mathrm{R},0}(t) = -i\, \theta(t)\, e^{-i\epsilon_k^{\mathrm{LDA}} t}. \tag{21}$$

Kas, Rehr, and Reining showed that, in the thermodynamic limit, the retarded cumulant yields a more physical spectral function because it treats particles and holes on an equal footing [1]. They also showed that, at the bottom of the band, the TC and RC are essentially equivalent.

D. Quasiboson formalism

The methods discussed below solve the Hamiltonian

$$\hat{H} = \hat{H}_{\mathrm{el}} + \hat{H}_{\mathrm{pl}} + \hat{H}_{\mathrm{ep}} \tag{22}$$

with $\hat{H}_{\mathrm{el}} = \sum_{\boldsymbol{k}\in\mathcal{O}} \epsilon_{\boldsymbol{k}}^{\mathrm{HF}} a_{\boldsymbol{k}}^\dagger a_{\boldsymbol{k}}$, $\hat{H}_{\mathrm{pl}} = \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} \omega_{\nu_{\boldsymbol{q}}} b_{\nu_{\boldsymbol{q}}}^\dagger b_{\nu_{\boldsymbol{q}}}$, and $\hat{H}_{\mathrm{ep}} = \sum_{\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}} \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q}) a_{\boldsymbol{k}+\boldsymbol{q}}^\dagger a_{\boldsymbol{k}} \left(b_{\nu_{-\boldsymbol{q}}}^\dagger + b_{\nu_{\boldsymbol{q}}}\right)$, where $\mathcal{O}$ and $\mathcal{M}$ are the sets of occupied bands and plasmonic modes, respectively. As indicated, given a plasmonic mode ν, the associated momentum transfer $\boldsymbol{q}$ is already determined. The Plasmonic frequencies $\omega_{\nu_{\boldsymbol{q}}}$and electron-plasmon couplings $g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})$ are The eigenvalues and transition densities taken from a direct RPA calculation starting from n. LDA reference as . The seasoned GW practitioner may be wary of this formulation because it seemingly mixes Hartree-fock single-particle energies with LDA screening. However, this has been done with empirical success in the self-consistent GW community for decades, so we pursue the same approach in defining the quasiboson Hamiltonian.

1. Self-consistent cumulant expansion

For many decades, self-consistency has been pursued has been pursued within the Dyson framework. There have only been a few studies of carrying out the self-consistency in the cumulant formalism, which have shown promise for computing spectral functions of the hosting model . This is the first. application of the self-consistent cumulant expansion to A fermionic system in the electron gas. Note that this has been done before by Holmes, but this is not true self-consistency, as we show in the appendix. Specifically, we compute the retarded Green's function in the time domain as In equation X, and then we solve for the cumulant as We are always solving

$$\frac{dC_k}{dt} = -i\Lambda_k - \sum_{i\in\mathcal{S}} \sum_{\nu} |g_{ki\nu}|^2 \int_0^t d\tau \; e^{-i\Delta_{ki\nu}(t-\tau)} \; e^{-[C_k(t)-C_k(\tau)]+[C_i(t)} \tag{23}$$

where $\Delta_{ki\nu} = \epsilon_i^{\mathrm{LDA}} - \epsilon_k^{\mathrm{LDA}} - \Omega_\nu$, Λ_k is a constant, $g_{ki\nu}$ is the electron-plasmon coupling, and $\mathcal{S}$ is some set of states.

2. Delocalized coupled-cluster theory

The D2 ansatz is We compute the DCC ground state by considering the following processes of excitation out of the reference Hartree-fock reference. Then we apply the tangent-space formalism to compute the spectral functions. We use the nomenclature dCC-X-SY-(Z, A) To denote a method, which, For the ground state calculation is X order In the fermions with Y coupled electron-boson excitations. See this paper for more details.

III. RESULTS

Here, we study finite supercells of the electron gas at rs = 4, converging onto the thermodynamic limit. To begin, we consider the smallest possible supercell with 14 electrons in 19 orbitals (14e19o), for which, We can compare the methods of our study with the numerically exact result with DMRG, which well coincides wtith CCSDT. For 114e485o,

IV. CONCLUSIONS

[1] J. J. Kas, J. J. Rehr, and L. Reining, Cumulant expansion of the retarded one-electron Green function, Phys. Rev. B **90**, 085112 (2014).